C3.1 Algebraic Topology

1100508

January 15, 2026

The project is presented in prose but we give a guide here as to where explicitely each question is answered:

- Question 1 – Chapter 1
- Question 2 – Chapter 1 and Chapter 2
- Question 3 – Section 2.1
- Question 4 – Section 2.1.1
- Question 5 – Section 2.2
- Question 6 – Chapter 3
- Question 7 – Chapter 4

Contents

Table 1: Summary of Work Done in this Project

Space	$\mathbf{H}_*$	$\mathbf{H}^*$	**Up to Hom Eq**	k
$\mathrm{Conf}_k(\mathbb{R}^n)$	✓		✓	2
$\mathrm{UConf}_k(\mathbb{R}^2)$	✓, $* = 1$		✓	> 0
$\mathrm{Conf}_k(S^1)$	✓		✓	> 0
$\mathrm{Sym}_k(S^2)$	✓	✓	✓	> 0
$\mathrm{Conf}_k(S^2)$	✓	✓	✓	$2, 3$
$\mathrm{UConf}_k(S^2)$	✓		✓	2
$\mathrm{Conf}_k(T^2)$	✓		✓	2

Table 2: Line denotes work outside the brief

1 Configuration Spaces of $\mathbb{R}^n$

Goal: Show $\mathrm{Conf}_2(\mathbb{R}^2) \cong \mathbb{R}^3 \times S^1$ and generalize.

The ordered configuration space of a topological space X is:

$$\mathrm{Conf}_n(X) := \{(x_1, \dots, x_n) \in X^n | x_i \neq x_j \text{ for } i \neq j\}$$

To examine the case when $X = \mathbb{R}^2$ we consider any two points in $x, y \in \mathbb{R}^2, x \neq y$.

Lemma 1. *We need exactly three quantities to uniquely determine a tuple of distinct points in $\mathbb{R}^2$:*

- *Euclidean distance:* $\|x - y\|_{\mathbb{R}^2}$
- *Midpoint or centre of mass:* $\frac{(x+y)}{2}$
- *Normalized Unit Vector of Line Between them:* $\frac{x-y}{\|x-y\|_{\mathbb{R}^2}}$

Proof. Consider the map

$$\begin{gathered} g : \mathrm{Conf}_2(\mathbb{R}^2) \to \mathbb{R}_{>0} \times \mathbb{R}^2 \times S^1 \\ (x, y) \mapsto (\|x - y\|_{\mathbb{R}^2}, \frac{x + y}{2}, \frac{x - y}{\|x - y\|_{\mathbb{R}^2}}) \end{gathered}$$

It suffices to show that this is a continuous map that is injective. Continuity follows from the fact that the maps into each component are continuous. For injectivity, suppose $(x, y) \neq (w, z)$ and suppose distance $d \in \mathbb{R}$ and centre of mass $m \in \mathbb{R}^2$ coincide. Then $y = 2m - x, z = 2m - w$ and thus the normalized unit vectors $\hat{xy} = \frac{2x-2m}{d}, \hat{wz} = \frac{2w-2m}{d}$. If $\hat{wz} = \hat{xy}$ then $x = w$ and thus $y = z$ which is a contradiction. $\square$

In fact the map g is surjective. The tuple $(d, m, \hat{u})$, with $d > 0$ lies in the pre-image of $((m + \frac{d}{2} \cdot \hat{u}), (m - \frac{d}{2} \cdot \hat{u}))$. We note that $\|x - y\|_{\mathbb{R}^2}$ spans $(0, \infty)$, the centre of mass $\frac{x+y}{2}$ spans $\mathbb{R}^2$ and the set of normalized unit vectors spans S^1.

Since g is bijective we show that g^{-1} is continuous as well, since the closed ball centered at m, $g(\bar{B}(m, \|x - y\|)) = (0, d] \times \bar{B}(m, \|x - y\|) \times S^1$ which is closed in the codomain.

Note that the way we defined the homeomorphism g is agnostic to the dimension of the space as long as we are dealing with two points, and in fact we can define a similar map:

$$g : \mathrm{Conf}_2(\mathbb{R}^n) \to \mathbb{R}_{>0} \times \mathbb{R}^n \times S^{n-1}$$
$$(x, y) \mapsto (\|x - y\|_{\mathbb{R}^n}, \frac{x + y}{2}, \frac{x - y}{\|x - y\|_{\mathbb{R}^n}})$$

Note that $\mathbb{R}_{>0}$ is homeomorphic to $\mathbb{R}$ via $\ln(x)$. Thus $\mathrm{Conf}_2(\mathbb{R}^n) \cong \mathbb{R}^{n+1} \times S^{n-1}$.

2 Homology Computation

Goal: Compute $H_*(\mathrm{Conf}_2(\mathbb{R}^n))$

We use the homeomorphism described in Lemma 1 to compute the homology using the algebraic Künneth formula. The isomorphism induced by g_* on homology groups implies $H_*(\mathrm{Conf}_2(\mathbb{R}^n)) \cong H_*(\mathbb{R}^{n+1} \times S^{n-1})$. Since $H_*(S^{n-1}; \mathbb{Z})$ is free abelian, all Tor terms vanish in the Künneth formula. Thus, for $k \geq 1$, the following chain of isomorphisms hold.

$$\begin{aligned} H_k(\mathbb{R}^{n+1} \times S^{n-1}; \mathbb{Z}) &\cong \bigoplus_{i+j=k} H_i(\mathbb{R}^{n+1}; \mathbb{Z}) \otimes H_j(S^{n-1}; \mathbb{Z}) \\ &\cong H_0(\mathbb{R}^{n+1}; \mathbb{Z}) \otimes H_k(S^{n-1}; \mathbb{Z}) \\ &\cong \mathbb{Z} \otimes H_k(S^{n-1}, \mathbb{Z}) \cong H_k(S^{n-1}, \mathbb{Z}) \end{aligned}$$

The big direct sum collapses due to $H_i(\mathbb{R}^{n+1}; \mathbb{Z}) = 0, \forall i > 0, \mathbb{Z}$ when $i = 0$, and using the identity $0 \otimes A = 0$. The last isomorphism the standard isomorphism $\mathbb{Z} \otimes A \cong A$, since $H_0(\mathbb{R}^{n+1}; \mathbb{Z}) \cong \mathbb{Z}$. Finally, since the space is path-connected (being the product of path-connected spaces), $H_0(\mathbb{R}^{n+1} \times S^{n-1}) = \mathbb{Z}$. Thus in totality,

$$H_k(\mathrm{Conf}_2(\mathbb{R}^n)) \cong H_k(S^{n-1}) = \begin{cases} \mathbb{Z}, & k = 0, n-1 \\ 0, & \text{otherwise} \end{cases}$$

2.1 Braid Groups

Goal: Compute $H_1(\mathrm{UConf}_n(\mathbb{R}^2))$

Definition. The unordered configuration space is defined as

$$\mathrm{UConf}_n(X) := \mathrm{Conf}_n(X)/S_n$$

where S_n is the *permutation group* on n elements.[Rol10]

To calculate $H_1(\mathrm{UConf}_n(\mathbb{R}^2))$ we examine the n-th *Braid Group* B_n, defined as $\pi_1(\mathrm{UConf}_n(\mathbb{C}))$. As a preliminary observation we note that $\mathbb{R}^2 \cong \mathbb{C}$. Moreover, by *Hurewicz Theorem* (Problem 8 Sheet 1) since $\mathbb{R}^2$ is path-connected, the *Hurewicz homomorphism* induces an isomorphism $\pi_1^{ab}(X) \cong H_1(X)$. Thus it suffices to compute the abelianization of B_n for $n \geq 2$. When $n = 1$, since S_1 is trivial, each point in $\mathbb{R}^2$ is in it's own equivalence class and thus the space $\mathrm{UConf}_1(\mathbb{R}^2) \cong \mathbb{R}^2$. Thus $H_1(\mathrm{UConf}_1(\mathbb{R}^2)) = 0$.

Suppose $n \geq 2$, then we choose the following presentation introduced by Artin [Art47] of B_n presented by the generators $\sigma_1, \dots, \sigma_{n-1}$ with the following relations:

$$\sigma_i\sigma_j = \sigma_j\sigma_i, |i-j| > 1 \tag{1}$$
$$\sigma_i\sigma_j\sigma_i = \sigma_j\sigma_i\sigma_j, |i-j| = 1 \tag{2}$$

Lemma 2. *The abelinization $B_n^{ab} := B_n/[B_n, B_n]$ is cyclic, where $[B_n, B_n]$ is the commutator of the group,*

Proof. By definition, quotienting by the commutator introduces the following relation amongst generators $\sigma_i\sigma_j = \sigma_j\sigma_i, \forall i \neq j$ in the presentation of B_n. Substituting these relations into (2) for $|i-j| = 1$.

$$\sigma_i^2\sigma_j = \sigma_j^2\sigma_i$$
$$\sigma_i^2 = \sigma_j\sigma_i$$
$$\sigma_i = \sigma_j$$

Hence, get a chain of equalities $\sigma_1 = \cdots = \sigma_{n-1}$ which implies that B_n^{ab} is generated by only one generator, say σ and is therefore cyclic.

$\square$

Lemma 3. $B_n^{ab} \cong \mathbb{Z}$

Proof. Define a map

$$\omega : B_n \longrightarrow \mathbb{Z}$$

by defining,

$$\omega(\sigma_i) = 1 \quad \text{for all } i = 1, \dots, n-1,$$

and extending multiplicatively, so that for a word $\sigma_{i_1}^{\alpha_1} \cdots \sigma_{i_k}^{\alpha_k}$ we have $\omega\left(\sigma_{i_1}^{\alpha_1} \cdots \sigma_{i_k}^{\alpha_k}\right) = \sum_{j=1}^k \alpha_j$ This map is well-defined. To see this any two generators are conjugate to each other,

$$(\sigma_k\sigma_{k-1})\sigma_k(\sigma_k\sigma_{k-1})^{-1} = (\sigma_{k-1}\sigma_k\sigma_{k-1})(\sigma_k\sigma_{k-1})^{-1} = \sigma_{k-1}$$

Moreover, $\omega(g\sigma_j g^{-1}) = \omega(\sigma_j), i \neq j$. Thus for any two identical expressions for a word $\beta, \bar{\beta}$, we may replace each occurance of a generator by the conjugate expression to σ_1.

$$0 = \omega(1_{B_n}) = \omega(\beta\bar{\beta}) = \omega\left((g_{i_1}\sigma_1 g_{i_1}^{-1})^{\alpha_1} \dots (\bar{g}_{i_1}\sigma_1 \bar{g}_{i_1}^{-1})^{-\bar{\alpha}'_k}\right) = \sum \alpha_i - \sum \bar{\alpha}_i$$

and so $\omega(\beta) = \omega(\bar{\beta})$. Moreover, ω is surjective since $\omega(\sigma_1) = 1$.

Since $\mathrm{Im}(\omega) = \mathbb{Z}$ is abelian, the commutator subgroup $[B_n, B_n]$ is contained in $\ker\omega$. Hence ω factors through the abelianization, yielding a surjective homomorphism.

$$\overline{\omega} : B_n/[B_n, B_n] \longrightarrow \mathbb{Z}.$$

Thus since B_n^{ab} is infinite and cyclic it is isomorphic to $\mathbb{Z}$. □

Theorem 4. $H_1(\mathit{UConf}_n(\mathbb{R}^2)) \cong \mathbb{Z}$

Proof. By Lemma 2.1 and the Hurewicz Theorem, it follows that

$$H_1(\mathrm{UConf}_n(\mathbb{R}^2)) \cong \pi_1^{ab}(\mathbb{C}) \cong B_n^{ab} \cong \mathbb{Z}$$

□

2.1.1 Homotopically Equivalent Subspaces

As a corollary consider $X = \mathbb{R}^2$, $Y = *$ (point space). Clearly for $n = 2$, since Y has only one distinct point, $\mathrm{UConf}_2(Y) = \emptyset$. Since X is contractible, it is homotopically equivalent to Y. But, by Lemma 2.1,

$$\mathbb{Z} \cong H_1(\mathrm{UConf}_2(X)) \ncong H_1(\mathrm{UConf}_2(Y)) \cong 0$$

2.2 Ordered Configurations of the 1-Sphere

Goal: Compute $H_*(\mathrm{Conf}_n(S^1))$

We examine the space $\mathrm{Conf}_n(S^1)$ in this subsection. Note that for $n = 0, 1$ the space is homeomorphic to a point and S^1 respectively and thus the homology groups are easy to describe. The interesting case occurs when $n \geq 2$. We constrain our analysis to S^1 as a subspace of $\mathbb{C}$, with the parameterization $z(t) \mapsto e^{it}$ where $z(t) \in S^1$. Looking at $\mathrm{Conf}_2(S^1)$ it is easy to construct a homeomorphism by considering the *angular* midpoint. A small caveat is that S^1 is orientable as a subspace and thus we must pick a direction to locate distinct points, say counter clock-wise. In this way, $z_1 \neq z_2 \in \mathrm{Conf}_2(S^1)$ are related as $z_1 = e^{it_1}$, $z_2 = e^{i(t_1+\theta)}$ for angle $\theta \in (0, 2\pi)$, otherwise there exists two antipodal midpoints on S^1.

Lemma 5. *d is a homeomorphism and $\mathit{Conf}_2(S^1) \cong S^1 \times (0, 2\pi)$.*

$$d : (e^{it_1}, e^{i(t_1+\theta)}) \mapsto (e^{i(\frac{t_1+\theta/2}{2})}, \theta)$$

Proof. □

Translating this same idea to $n \geq 3$ is straightforward. The intuition is that we can "pin" the first point z_1 and describe the remaining $n-1$ points by their angular offsets relative to z_1. Since the points are distinct, none of these offsets can be 0 or integer multiples of 2π. If we write $z_1 = e^{it}$ and $z_k = e^{i(t+u_{k-1})}$ for relative angles $u_j \in (0, 2\pi)$, we consider the map d' which sends a configuration to the position of the first point and the relative coordinates of the subsequent points:

$$d' : (e^{it}, e^{i(t+u_1)}, \ldots, e^{i(t+u_{n-1})}) \mapsto (e^{it}, (u_1, \ldots, u_{n-1}))$$

The first component lies in S^1. The remaining tuple $(u_1, \ldots, u_{n-1})$ represents $n-1$ distinct points in the interval $(0, 2\pi)$ (since $z_i \neq z_j$ implies $u_{i-1} \neq u_{j-1}$). Thus, the image of the relative coordinates lies in the configuration space of the interval. This yields the generalized homeomorphism:

$$\mathrm{Conf}_n(S^1) \cong S^1 \times \mathrm{Conf}_{n-1}((0, 2\pi))$$

Since $(0, 2\pi)$ is homeomorphic to the standard interval $I = (0, 1)$, this effectively decomposes the circular configuration space into a circle and the linear configuration space of $n-1$ points.

2.2.1 Configuration Space of the Interval

The space $\mathrm{Conf}_n(I)$ for an interval $I \subset \mathbb{R}$ can be imagined as follows.
Consider the n points $(p_1, \ldots, p_n)$ on a subset of the real line. They have a natural ordering $p_{z(1)} < p_{z(2)} < \cdots < p_{z(n)}$ for some $z \in S_n$, and each since point is 'tagged' with which copy of the interval it came from, only points preserving the ordering are path-connected. To see this any path with a single inversion (p_i, p_j) from z has to cross the point with $p_i = p_j$ which is forbidden in the configuration space. Thus the set,

$$C_z = \{(p_1, \ldots, p_n) | p_{z(1)} < p_{z(2)} < \cdots < p_{z(n)}, z \in S_n\}$$

corresponds to a path-component. Since each C_z is a product of contractible segments $(p_1, p_2) \subset \mathbb{R}$ it is by itself contractible.

$$\mathrm{Conf}_n(I) = \bigsqcup_{z \in S_n} C_z$$

As $|S_n| = n!$, $\mathrm{Conf}_n(I)$ is a disjoint union of $n!$ contractible path-components. Next, the homology of such a space is easy to compute.

$$H_k(\mathrm{Conf}_n(I); \mathbb{Z}) = H_k(\bigsqcup_{z \in S_n} C_z, \mathbb{Z}) = \begin{cases} \mathbb{Z}^{n!}, & k = 0 \\ 0, & \text{otherwise} \end{cases}$$

2.2.2 Homology of $Conf_n(S^1)$

The homology can again be calculated using the algebraic Künneth formula. Let $I = (0, 2\pi)$. Again as all homology groups involved are free abelian; hence the Tor term is zero in the Künneth formula. Let $n > 1$.

$$\begin{aligned} H_k(S^1 \times \mathrm{Conf}_n(I); \mathbb{Z}) &\cong \bigoplus_{i+j=k} H_i(S^1; \mathbb{Z}) \otimes H_j(\mathrm{Conf}_{n-1}(I); \mathbb{Z}) \\ &= \begin{cases} \mathbb{Z}^{(n-1)!}, & k = 0 \\ \mathbb{Z}^{(n-1)!}, & k = 1 \\ 0, & \text{otherwise} \end{cases} \end{aligned}$$

This follows as for $k = 1$, we are left with $H_1(S^1; \mathbb{Z}) \otimes H_0(\mathrm{Conf}_{n-1}(I); \mathbb{Z}) \oplus 0 \cong \mathbb{Z} \otimes \mathbb{Z}^{(n-1)!} \cong \mathbb{Z}^{(n-1)!}$. For $k > 1$ at least one of the factors are zero in each tensor product.

3 Symmetric Product of Manifolds

Goal: Prove that $\mathrm{Sym}^n(X)$ for a closed, connected, orientable surface X is a manifold.

We analyse the following *symmetric product space* of X denoted $\mathrm{Sym}^n(X)$

Definition. $\mathrm{Sym}^n(X) := X^n/S_n$ for a topological space X.

A n-dimensional manifold is defined as follows.

Definition. A n-dimensional manifold M is a Hausdorff, second countable (countable base) topological space such that

- $\forall p \in M, \exists U_p$ open in M such that $p \in U_p$ and $U_p \cong V \subset \mathbb{R}^n$ where V is open.

A *surface* is a 2-dimensional manifold. A *compact* manifold is called *closed.*

We adapt the definition of an orientable manifold from [Hat02] (pg 234).

Remark 1. A crucial fact, that we use without proof is that once we pick an orientation, $\mathbb{R}^2$ is locally (around a point) homeomorphic to $\mathbb{C}$.

We also use the following theorem without proof from group actions on spaces.

Theorem 6 (Quotients by finite group actions [Har+23])**.** *Let G be a finite group acting continuously on a Hausdorff space Y, and let $U \subset Y$ be an open G–invariant subset. Then:*

1. *The induced map $U/G \to Y/G$ is a homeomorphism onto its image.*

2. *If $U = U_1 \times \cdots \times U_k$ and the action of G on U splits as a product action $G_1 \times \cdots \times G_k$, then there is a natural homeomorphism*

$$U/G \;\cong\; (U_1/G_1) \times \cdots \times (U_k/G_k).$$

[Hat02](Consequence of Chapter 1.3 Exercise 22)

Theorem 7. *Let X be a closed, connected, orientable surface then $Sym^n(X)$ is a manifold.*

In order to prove this theorem we note that the group S_n acts on X^n by the following:

$$\sigma(x_1, \ldots, x_n) = (x_{\sigma(1)}, \ldots, x_{\sigma(n)}), \sigma \in S_n$$

For a point $x = (x_1, \ldots, x_n) \in X^n$ we refer to the orbit of the action on $x \in Sym^n(X)$ as $[x_1, \ldots, x_n]$ induced by the canonical quotient map $q : X^n \to \mathrm{Sym}^n(X)$.

Our first task is to show how to pass from open subsets in X^n to open subsets in $\mathrm{Sym}^n(X)$.

Consider the point $[x_1, \ldots, x_n] \in \mathrm{Sym}^n(X)$. Suppose there are $k \leq n$ distinct points in this collection, since all permutations are identified we can group by the multiplicity of each occurance of $x_i \in X$ by writing.

$$[x_1, \ldots, x_n] = m_1 \cdot y_1 + \cdots + m_k \cdot y_k, \sum_{i=1}^{k} m_i = n \text{ and } y_i \in X^n \text{ distinct}$$

Since X is Hausdorff we can find pairwise disjoint neighbourhoods V_i such that $V_1^{m_1} \times \cdots \times V_k^{m_k}$ open in the ordered space X^n and each contains exactly one distinct point $y_i \in V_i$.

Now let us collect the orbits of the action of S_n on an open set U,

$$S_n \cdot U := \bigcup_{x \in U} S_n \cdot x$$

Lemma 8. *If $U \in X^n$ is open and $S_n \cdot U = U$ then $q(U)$ is open in $Sym^n(X)$.*

Proof. Recall that q is a quotient map, so a set $O \in \mathrm{Sym}^n(X)$ is open iff $q^{-1}(O)$ is open in X^n. Now $q^{-1}(q(U))$ is precisely equal to $S_n \cdot U = U$ and is therefore open. Moreover, $q^{-1}(q(U))$ open implies $q(U)$ open since q is a quotient map. $\square$

Lemma 9. *q sends $W = V_1^{m_1} \times \cdots \times V_k^{m_k}$ to $q(W)$ open in $Sym^n(X)$ where $q(W) \cong Sym^{m_1}(V_1) \times \cdots \times Sym^{m_k}(V_k)$.*

Proof. Consider again the set of all orbits of W notated by elements of S_n instead,

$$\hat{W} = S_n \cdot W := \bigcup_{\sigma \in S_n} \sigma(W)$$

By construction, $S_n \cdot \hat{W} = S_n \cdot (S_n \cdot W) = S_n \cdot W = \hat{W}$, and since each $\sigma(W)$ (permutation of product of open sets) is open in X^n their union is also open.

By the Lemma 8 above $q(\hat{W})$ is open. Since we mod out the group action in the quotient space, $q(\hat{W}) = q(W)$.

We have shown that $\hat{W}$ is S_n-invariant by showing $S_n \cdot \hat{W} = \hat{W}$. It is also open and so is $q(W) = q(\hat{W})$. Moreover, because the sets $V_1, \dots, V_k$ are pairwise disjoint, the action of S_n on $\hat{W}$ splits as the product action of $S_{m_1} \times \cdots \times S_{m_k}$ on $V_1^{m_1} \times \cdots \times V_k^{m_k}$. By Theorem 6, it follows that

$$q(W) \;\cong\; \mathrm{Sym}^{m_1}(V_1) \times \cdots \times \mathrm{Sym}^{m_k}(V_k),$$

and this set is open in $\mathrm{Sym}^n(X)$. □

Proof. (Theorem 7) Since $\mathrm{Sym}^{m_1}(V_1) \times \cdots \times \mathrm{Sym}^{m_k}(V_k)$ with $m_1 + \cdots + m_k = n$ is a neighbourhood of $[x_1, \dots, x_n] \in \mathrm{Sym}^n(X)$ by construction and $x_i = x_j$ with $x_i \in V_i$ then $x_i, x_j \in V_i$, Lemma 9 allows us to restrict our attention to $\mathrm{Sym}^{m_i}(V_i)$. To conclude that the space is a manifold, what remains to show is that each m_i, $\mathrm{Sym}^{m_i}(V_i) \cong U$ open in $\mathbb{C}^{m_i}$. For the remainder of this subsection let $m_i = m$.

Since X is a surface, for each V_i there is a homeomorphism $\phi_i : V_i \to \mathbb{C}$. The induced map $\bar{\phi}$ respects the quotient by S_n and is therefore a homeomorphism. Consider the complex monic polynomial.

$$p_w(z) = \prod_{j=1}^{m} (z - w_j) = z^m + a_m z^{m-1} + \cdots + a_2 z + a_1$$

To each point $w = [w_1, \dots, w_m]$ (where w_i need not be distinct) in the unordered complex space $\mathrm{Sym}^m(\mathbb{C})$ we can associate the coefficients of the polynomial $p_w(z)$ defined by the map:

$$F : \mathrm{Sym}^m(\mathbb{C}) \to \mathbb{C}^m$$
$$F(w) \mapsto (a_1, \dots, a_m)$$

This is well-defined since any two points are equivalent up to reordering and thus correspond to the same coefficients. Moreover,F is a homeomorphism. Recall that $\mathbb{C}[z]$ is a UFD (unique factorization domain).

1. **Injective:** Suppose $F(w) = F(z)$ then since the coefficients are equal, the polynomials p_w, p_z coincide. Since $\mathbb{C}[x]$ is a UFD the roots of the monic polynomials p_w, p_z up to reordering and including multiplicities also coincide. Thus $w = z$ in $\mathrm{Sym}^m(\mathbb{C})$.

2. **Surjective:** Let $p(z) = z^m + a_m z^{m-1} + \cdots + a_2 z + a_1$ correspond to $(a_1, \dots, a_m)$. Since the degree of the monic polynomial p is m, by the Fundamental Theorem of Algebra this polynomial factors as a product of exactly m linear factors over $\mathbb{C}$ counted with multiplicity. Thus $p(z) = \prod_{j=1}^{m} (z - w_j)$ which is the image of $[w_1, \dots, w_m]$.

3. **Continuity and Continuous Inverse:** Continuity of the inverse follows from the classical theorem that the roots of a complex polynomial depend continuously on its coefficients when considered as an unordered multiset, (see [NR23]).

The required homeomorphism is

$$\mathrm{Sym}^m(V_i) \xrightarrow{\bar{\phi}_i} \mathrm{Sym}^m(\mathbb{C}) \xrightarrow{F} \mathbb{C}^m$$

and thus $\mathrm{Sym}^m(V_i) \cong \mathbb{C}^m$. We can extend this homeomorphism to the finite product as well.

$$\mathrm{Sym}^{m_1}(V_1) \times \cdots \times \mathrm{Sym}^{m_k}(V_k) \cong \mathbb{C}^{m_1} \times \cdots \times \mathbb{C}^{m_k}$$

The dimensions of the euclidean space is $2m_1 + \cdots + 2m_k = 2n$ since $\mathbb{C} \cong \mathbb{R}^2$. Thus the space $\mathrm{Sym}^n(X)$ is a $2n$ dimensional manifold.

□

4 Other Constructions on Orientable, Closed, Manifolds

4.1 $Sym^n(S^2)$

Goal: Give a complete characterisation of $Sym^n(S^2)$.

We provide a brief discussion on the result that $Sym^n(S^2) \cong \mathbb{C}P^n$. A large portion of the discussion can be inferred from discussions in the lecture notes.

We can view S^2 as the one-point compactification of the complex plane, $\mathbb{C} \cup \{\infty\} = \mathbb{C}P^1$. As a result, $S^2 \cong \mathbb{C}P^1$. Note that $Sym^n(S^2) = S^{2n}/S_n$ and that

$$\mathbb{C}P^n = \{[z] = [z_0 : \ldots : z_n] : \text{ not all } z_j = 0 \in \mathbb{C}, [z] \sim [\lambda z]\}$$

Consider a homogenous bi-variate polynomial $q(W, Z)$.

$$q : \mathbb{C} \times \mathbb{C} \to \mathbb{C}$$

$$q(W, Z) = \prod_{i=1}^{n}(w_i Z - z_i W) = a_n Z^n + a_{n-1} Z^{n-1} W + \cdots + a_0 W^n$$

The roots $[w_i : z_i]$ of such a polynomial are unordered and can be identified as points in $Sym^n(S^2) = Sym^n(\mathbb{C}P^1)$. Meanwhile, scaling the $(n+1)$ tuple of coefficients of this polynomial does not scale the roots and are therefore determined by the roots. This fact lends itself to a well-defined homeomorphism.

$$\Phi : Sym^n(\mathbb{C}P^1) \to \mathbb{C}P^n$$
$$[[w_1 : z_1], \ldots, [w_n : z_n]] \mapsto [a_0 : \ldots : a_n]$$

The point at infinity (let it be $[1 : 0]$) forces the coefficient a_n to be zero and thus reduces the degree to less than n, but $[a_1 : a_2 \ldots : 0]$ is a valid point in $\mathbb{C}P^n$ and the map remains well-defined. By an earlier discussion roots vary continuously with coefficients in projective sense as well. By well established theorems in algebraic geometry, a homogenous polynomial in $\mathbb{C}[x, y]$ with degree n has at most n projective nontrivial root $[w_i : z_i] \neq [0 : 0]$, which are perfectly valid elements of $Sym^n(S^2)$. Finally, injectivity follows from the fact that the roots not only define the coefficients up to rescaling but do so *uniquely*.

4.1.1 Homology and Cohomology of $\mathbb{C}P^n$

> **Goal:** Develop the cohomology ring of $\mathbb{C}P^n$ using *Poincaré* duality.

This draws from the lecture notes. CP^n as a CW-complex has a single cell in even dimensions, $e^0, e^2, \ldots, e^{2n}$. The cellular differentials are therefore all zero. Thus,

$$C_n^{CW}(\mathbb{C}P^n) = \begin{cases} \mathbb{Z}e^k & k = 0, 2, \ldots, 2n \\ 0 & \text{otherwise} \end{cases}$$

$$H_*(\mathrm{Sym}^n(S^2)) \cong H_*^{CW}(\mathbb{C}P^n) = \begin{cases} \mathbb{Z} & * = 0, 2, \ldots, 2n \\ 0 & \text{otherwise} \end{cases}$$

moreover, by the *Universal Coefficients Theorem* for $k \leq n$

$$H^{2k}(\mathbb{C}P^n; \mathbb{Z}) = Hom(H_{2k}(\mathbb{C}P^n), \mathbb{Z}) \oplus \mathrm{Ext}(H_{2k-1}(\mathbb{C}P^n), \mathbb{Z}) \cong Hom(\mathbb{Z}, \mathbb{Z}) \oplus 0 \cong \mathbb{Z}$$
$$H^{2k-1}(\mathbb{C}P^n; \mathbb{Z}) = Hom(H_{2k-1}(\mathbb{C}P^n), \mathbb{Z}) \oplus \mathrm{Ext}(H_{2k}(\mathbb{C}P^n), \mathbb{Z}) \cong Hom(0, \mathbb{Z}) \oplus 0 \cong 0$$

There are two ways to describe the cohomology ring of $\mathbb{C}P^n$ over $\mathbb{Z}$ which by itself is an important note. The first is by using the Künneth formula and the naturality of the cup product which was covered in Sheet 3 Exercise 4. We present an alternate approach by appealing to *Poincaré Duality* found in [Hat02]. By the classification of finitely generated abelian groups, (co)homology groups have the structure of $\mathbb{Z}^r \oplus$ (torsion). *Poincaré Duality* asserts the existence of an isomorphism $D : H^k(M; R) \to H_{n-k}(M; R)$ for a closed, orientable manifold M. D sends an element in the cohomology α to the *cap product*, $[M] \frown \alpha$, where $[M] \in H_{n-k}(M; R)$ is the fundamental class of M. Such a class always exists for a closed connected and orientable manifold. The relationship between the cup and cap product is as follows:

$$\gamma(\alpha \frown \phi) = (\phi \smile \gamma)(\alpha)$$

at the chain and cochain level. Applying the cohomology and homology functors, makes the following diagram commute.

$$\begin{tikzcd} H^l(X; R) \arrow[r, "h_1"'] \arrow[d, "\phi\smile"] & Hom_R(H_l(X; R), R) \arrow[d, "(\frown\phi)^*"'] \\ H^{k+l}(X; R) \arrow[r, "h_2"] & Hom_R(H_l(X; R), R) \end{tikzcd}$$

The h_1, h_2 are from the *Universal Coefficients Theorem.* In our case, since the homology groups of $\mathbb{C}P^n$ are free when coefficients are taken in $R = \mathbb{Z}$, the maps h_1, h_2 are isomorphisms.

Thus, $(\phi \smile)$ is the dual of $(\frown \phi)$. Furthermore, for a closed, orientable n dimensional manifold the cup product pairing map

$$H^k(M;R) \times H^{n-k}(M;R) \to R, (\phi, \gamma) \mapsto (\phi \smile \gamma)[M]$$

is called *nonsingular* if the *adjoint* maps $d_1 : (H^k(M;R) \to Hom_R(H^{n-k}(M;R), R)), d_2 : (H^{n-k}(M;R) \to Hom_R(H^k(M;R), R))$ are both isomorphisms.

Lemma 10. *[Hat02] d_1, d_2 are isomorphisms for $R = \mathbb{Z}$ and $M = \mathbb{C}P^n$*

Proof. Let the dual of the Poincaré duality map formed by applying the contravariant functor $\mathrm{Hom}(--, R)$ be D^*. $D^* \circ h_2$ sends an element of the cochain $\gamma \in H^{n-k}(M;R)$ to the homomorphism $(\phi \mapsto \gamma([M] \frown \phi)) = (\gamma \smile \phi)[M])$. Since h_2 and D are both isomorphisms the composition $d_2 = D^* \circ h_2$ is also an isomorphism. The same holds for d_1 using the fact that the cup product is graded commutative. □

A consequence of nonsingularity is the following lemma that we state without proof, which says we can push the free part of a generator for the kth cohomology group to find a generator of the nth cohomology group. [Hat02]

Lemma 11. *[Hat02] If M is a closed, connected orientable n dimensional manifold, then an element $\alpha \in H^k(M, \mathbb{Z})$ generates an infinite cyclic summand $\iff$ there exists $\beta \in H^{n-k}(M, \mathbb{Z})$ such that $\alpha \smile \beta$ is a generator for $H^n(M, \mathbb{Z}) \cong \mathbb{Z}$.*

From this the cohomology ring can be easily uncovered.

Theorem 12. $H^*(\mathbb{C}P^n, \mathbb{Z}) \cong \mathbb{Z}[\alpha]/(\alpha^{n+1})$ *with* $\deg(\alpha) = 2$.

Proof. S^2 is a closed, orientable manifold and therefore, by the previous section, $\mathrm{Sym}^n(S^2)$ is a manifold as well. Since it is homeomorphic to $\mathbb{C}P^n$, it is closed and orientable for any $n \geq 1$ as well. Since $\mathbb{C}P^{n-1} \subset \mathbb{C}P^n$, by the LES of the pair $(\mathbb{C}P^{n-1}, \mathbb{C}P^n)$ gives us isomorphisms for all lower homology groups. That is, $H^i(\mathbb{C}P^n;\mathbb{Z}) \cong H^i(\mathbb{C}P^{n-1};\mathbb{Z})$ for $i \leq 2(n-1)$. By induction on n, suppose $H^*(\mathbb{C}P^{n-1}, \mathbb{Z}) \cong \mathbb{Z}[\alpha]/(\alpha^n)$ and thus $H^{2i}(\mathbb{C}P^n;\mathbb{Z})$ is generated by α^i for $i < n$.

Since the $2i$ level cohomology is infinite and cyclic, its generator α^i satisfies the hypotheses of Lemma 11. Fixing $i = 1$, and applying Lemma 11, there exists an element $\beta \in H^{2(n-1)}(\mathbb{C}P^n, \mathbb{Z})$ such that $\alpha \smile \beta$ generates $H^{2n}(\mathbb{C}P^n, \mathbb{Z})$. Since α^{n-1} generates the $2(n-1)$ cohomology which is isomorphic to $\mathbb{Z}$, β has the form $m\alpha^{n-1}$.

Since $H^{2n}(\mathbb{C}P^n;\mathbb{Z}) \cong \mathbb{Z}$, any isomorphism $H^{2n}(\mathbb{C}P^n;\mathbb{Z}) \to \mathbb{Z}$ sends a generator to ± 1. Hence $\alpha \smile \beta = \alpha \smile (m\alpha^{n-1}) = m\alpha^n$ is a generator of $H^{2n}(\mathbb{C}P^n;\mathbb{Z})$ if and only if $m = \pm 1$, and the result follows.

□

4.2 Conf$_n(S^2)$ for $n = 2, 3$

Goal: Show $\mathrm{Conf}_3(S^2) \cong S^2$ and $\mathrm{Conf}_3(S^2) \cong \mathbb{R}P^3$ where $\cong$ is interpreted as homotopy equivalences between the two spaces.

$\mathrm{Conf}_n(X)$ can be thought of as X^n with the big diagonal $\Delta = \{(x_1, \ldots, x_n) \in X^n : x_i = x_j \text{ for some } i \neq j\}$ removed. Focusing on S^2, consider the continuous and surjective projection $p : \mathrm{Conf}_2(S^2) \to S^2$ that sends $(x, y) \in S^2 \times S^2 \setminus \Delta \mapsto x$.The *fiber* of the projection $p^{-1}(b) = S^2 - \{b\} \cong \mathbb{R}^2$, which is a contractible space. Since S^2 is a 2 dimensional manifold, for every $x \in S^2$ we can pick an open neighbourhood $U_x \subset S^2$ that is an open hemisphere bounded by the plane through the origin. We define the map

$$h_x : p^{-1}(U_x) \to U_x \times H \cong U_x \times \mathbb{R}^2$$

as

$$h_x((y, z)) = \left(y, \frac{z - y}{\|z - y\|}\right)$$

where $H \subset S^2$ is the open hemisphere of unit direction vectors in $\mathbb{R}^3$ that point "away" from y. For a fixed $y \in U_x$, the map $f_y(z) = \frac{z-y}{\|z-y\|}$ provides a homeomorphism between the punctured sphere $S^2 \setminus \{y\}$ and the open hemisphere H. Since the vector subtraction and normalization depend continuously on the choice of y and z (given $y \neq z$), the map h_x is a homeomorphism over U_x. This demonstrates that p satisfies the local triviality property. Thus, $p : \mathrm{Conf}_2(S^2) \to S^2$ is a fiber bundle with contractible fiber $\mathbb{R}^2$ [Hat03].

It is at this point we must invoke a theorem that we give a proof sketch for since it lies outside the material.

Theorem 13. *The locally trivial fiber bundle p with contractible fibers induces a homotopy equivalence.*

Proof Sketch. Both the total space, $\mathrm{Conf}_2(S^2)$ and the base S^2 admit a CW-complex structure [FH01]. Therefore, by Whitehead's Theorem (applied to the CW-complexes) [Hat02] (Chapter 4), the long exact sequence associated with homotopy groups yields that p is a homotopy equivalence.[FH01] □

Thus $\mathrm{Conf}_2(S^2)$ is homotopically equivalent to S^2.

What we have deduced in the case $n = 2$ generalises to a description of the fibers known as the *Fadell-Niuwirth fibration* for the projection fiber bundle for larger n. This is the genesis construction in the study of configuration spaces.

Theorem 14. *[FN62](Theorem 3) For a fixed manifold M the map*

$$\pi : Conf_n(M) \to Conf_{n-1}(M)$$
$$\pi(x_1, \ldots, x_n) = (x_1, \ldots, x_{n-1})$$

is a locally trivial fiber bundle with fiber $Conf_1(M \setminus \{p_1, \ldots, p_{n-1}\})$ *homeomorphic to* $M \setminus \{p_1, \ldots, p_{n-1}\}$.

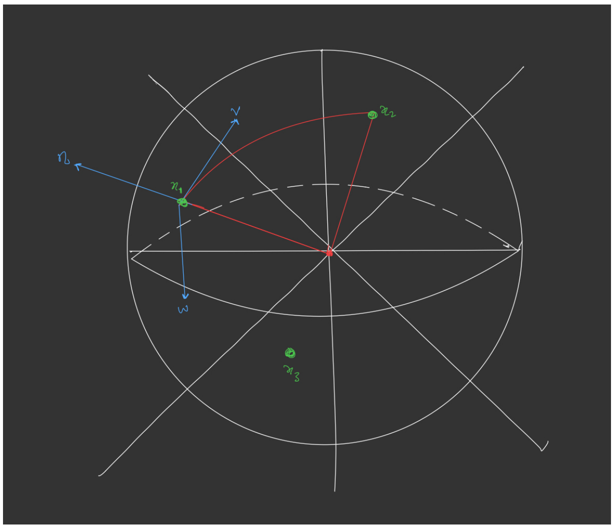

Figure 1: $w = n \times v$ and we pick either w or $-w$ depending on which side of the half-plane x_3 is on.

Consider the case of $n = 3$. $\mathrm{Conf}_3(S^2)$, which has fiber $S^2 - \{p_1, p_2\}$ is homotopically equivalent to S^1 and is therefore not contractible. The techniques used with $n = 2$ no longer apply to this case. However, the space still has structure as a locally trivial fiber bundle. Instead, we can give a differential topology interpretation of the space as described in [BT82].

For a orientable and smooth n manifold M, the *orthonormal frame* at a point x is an n dimensional orthonormal basis $\{e_1, \ldots, e_n\}$ of the tangent space at x, T_xM. The collection of all such orthonormal frames at all points forms another smooth manifold, called the frame bundle $F(M)$ with the associated projection:

$$p : F(M) \to M$$
$$p(x, e_1, \ldots, e_n) \mapsto x$$

The fiber p^{-1} consists of all bases of T_xM. If the manifold is orientable then the space oriented frame bundle $F^o(M)$ which is diffeomorphic to $SO(3)$, the special orthogonal group of 3×3 rotation matrices. [1]

Take $S^2 \subset \mathbb{R}^3$. The three distinct points on the circle $(x_1, x_2, x_3) \in \mathrm{Conf}_3(S^2)$ *uniquely* determine a oriented frame in $\mathbb{R}^3$. This is shown in Figure 1.

In this way we can map $\mathrm{Conf}_3(S^2)$ to the oriented frame bundle $F^o(S^2)$. Thus, $\mathrm{Conf}_3(S^2)$ is homotopically equivalent to $SO(3)$ [BT82][Hat03]. Note here we are not contradicting the famous 'Hairy Ball theorem'. The theorem says we can't pick a continuous non-zero field of tangent vectors on S^2 at each point $x \in S^2$. However, in our case, the choice of the vectors in T_xS^2 depends not just on one point x_1 but on the entire ordered configuration of three distinct points (x_1, x_2, x_3).

[1] Another viewpoint comes from using black-box arguments from [FH01]. We can use Proposition 1.1 which gives the map $O_{3,2} \to \mathrm{Conf}_3(S^2)$ is a homotopy equivalence. $O_{3,2}$ is the *Steifel Manifold* homotopically equivalent to $SO(3)$ using the same language of frames [Hat02].

n	$H_n(\mathrm{Conf}_3(S^2))$	$H^n(\mathrm{Conf}_3(S^2))$	$H_n(\mathrm{Conf}_2(S^2))$	$H^n(\mathrm{Conf}_2(S^2))$
0	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$
1	$\mathbb{Z}_2$	0	0	0
2	0	$\mathbb{Z}_2$	$\mathbb{Z}$	$\mathbb{Z}$
3	$\mathbb{Z}$	$\mathbb{Z}$	0	0
> 3	0	0	0	0

Lemma 15. $SO(3) \cong \mathbb{R}P^3$

We fix an orientation such that anticlockwise rotations around a fixed axis are considered positive. One interpretation of $\mathbb{R}P^3$, discussed in the lectures, was the disk D^3 with antipodal points on the boundary $\partial D^3 = S^2$ identified.

Proof. Let $R_\pi^x \in SO(3)$ be the rotation matrix by π about the direction of the vector of a nonzero point x. The orientation preserving map $\gamma : D^3 \to SO(3)$ with action $(x \mapsto R_\pi^x x)$ on nonzero x and $0 \mapsto I_3$ is continuous. This map sends antipodal points to antipodal points along the boundary and thus the map

$$\overline{\gamma} : D^3/\{x \sim -x : x \in S^2\} = \mathbb{R}P^3 \to SO(3)$$

is a homeomorphism. It is surjective by definition of $SO(3)$ and injective since any two rotations by a fixed angle in $\mathbb{R}^3$ can only coincide if and only if the axes (the point a rotation fixes) are also the same. □

In totality, $\mathrm{Conf}_3(S^2)$ is homotopically equivalent to $\mathbb{R}P^3$.

The homology and cohomology groups are given in the table below using known results from the lectures.

4.3 UConf$_2(S^2)$

Goal: Show $\mathrm{UConf}_2(S^2) \cong \mathbb{R}P^2$ where $\cong$ is interpreted as homotopy equivalences between the two spaces.

We obtain the nth unordered configuration space by quotienting the ordered nth configuration space by S_n. That is,

$$\mathrm{UConf}_2(S^2) : ((S^2 \times S^2) \setminus \Delta)/S_2 = \mathrm{Conf}_2(S^2)/S_2$$

One viewpoint we had adopted as part of proving Theorem 7 was to consider the group action of S_n on a space X.

The map,

$$F : \mathrm{Conf}_2(S^2) \to S^2$$
$$F(x, y) = ((x - y)/||x - y||)$$

This map is well defined and continuous since $||x - y|| \neq 0$ as $x \neq y$. For any $v \in S^2$, the fiber $F^{-1}(v)$ consists of pairs (x, y) where y is the reflection of x across the plane perpendicular to v, subject to the condition $x \cdot v > 0$. Thus, the fiber is diffeomorphic to the open hemisphere:

$$F^{-1}(v) \cong \{x \in S^2 \mid x \cdot v > 0\} \cong \mathbb{R}^2$$

and therefore contractible. Moreover, $F(x, y) = ((x - y)/||x - y||) = -((y - x)/||x - y||) = -F(y, x)$. The group defined by the relation $(x, y) \sim (y, x)$ is isomorphic to $\mathbb{Z}_2$. Let the quotient map,

$$\overline{F} : \mathrm{UConf}_2(S^2) \to S^2/\mathbb{Z}_2 \cong \mathbb{R}P^2$$
$$\overline{F}(x, y) = [F(x, y)]$$

Where the $[F(x, y)]$ is the equivalence class $\{F(x, y), -F(x, y)\}$. Now consider $(x, y) \sim (y, x)$,

$$\overline{F}(x, y) = \{F(x, y), -F(x, y)\} = \{-F(y, x), F(x, y)\} = \overline{F}(y, x)$$

thus the map is well-defined. Using our previous analysis $\overline{F}$ is also a fiber bundle with contractible fibers and is thus a homotopy equivalence. [Hat03]

$$\mathrm{UConf}_2(S^2) = \mathrm{Conf}_2(S^2)/\mathbb{Z}_{\neq} \to S^2/\mathbb{Z}_2 \cong \mathbb{R}P^2$$

Notice that $\mathrm{UConf}_2(S^2)$ is therefore not orientable. [2]

4.4 Conf$_2(T^2)$

[2] The author acknowledges the existence of the Napolitano Cell complex for general n for computing the homology and cohomology groups as in [NAP03]. However this method proved to be extremely tedious and quite difficult to present.

> **Goal:** Show $\mathrm{Conf}_2(T^2) \cong T^2 \times (S^1 \vee S^1)$ where $\cong$ is interpreted as homotopy equivalences between the two spaces.

We focus now on a different space with more complex homology groups. While constructing the local trivialisation of the fiber bundle for $\mathrm{Conf}_2(S^2)$ the map $(x, y) \to (x, (x-y)/|x-y|)$ was a local homeomorphism. However, since S^2 does not admit a topological group structure, and thus addition is not well-defined globally, the map doesn't give a homeomorphism overall. This is not a problem we face with T^2 as it is in fact a topological group (Lie Group) with well defined addition. [BT82]

The map

$$M : \mathrm{Conf}_2(T^2) \to T^2 \times (T^2 - \{0\})$$
$$M(x, y) \mapsto (x, x - y)$$

is a homeomorphism, since $x \neq y \implies x - y \neq 0$ and by since subtraction is a continuous function on T^2, $M^{-1}(x, y) = (x, x - y)$ is continuous as well.

$T^2 - \{0\}$ is homotopically equivalent to the wedge of two circles $S^1 \vee S^1$. Thus, $\mathrm{Conf}_2(T^2) \cong T^2 \times S^1 \vee S^1$. Next, we calculate the homology using the algebraic Künneth formula and present the *Betti Numbers* (the ranks of the homology groups) since the homology groups are more interesting. We use the formula, $H_*(\vee_\alpha X_\alpha) = \bigoplus_\alpha H_*(X_\alpha)$. Recall that $H_j(T^2; \mathbb{Z}) \cong \mathbb{Z}$ for $j = 0, 2$, $H_1(T^2; \mathbb{Z}) \cong \mathbb{Z} \oplus \mathbb{Z}$ and 0 otherwise. The homology of $S^1 \vee S^1$ is free and thus the Tor terms vanish.

$$H_k(T^2 \times (S^1 \vee S^1); \mathbb{Z}) \cong \bigoplus_{i+j=k} H_i(T^2; \mathbb{Z}) \otimes H_j(S^1 \vee S^1)$$

Note that $H_0(S^1 \vee S^1) = \mathbb{Z}$ as it is path-connected otherwise it splits according to the wedge sum formula.

n	$H_n(\mathrm{Conf}_2(T^2))$	β
0	$\mathbb{Z}$	1
1	$\mathbb{Z}^2 \oplus \mathbb{Z}^2$	4
2	$\mathbb{Z}^4 \oplus \mathbb{Z}$	5
3	$\mathbb{Z}^2$	2
> 3	0	0

5 Conclusion

In this project we have thoroughly investigated ordered and unordered configuration spaces as well as symmetric products for various manifolds in low-dimensions. We examine various big-picture techniques and constructions like *Fiber bundles*, the *Fadell-Neuwirth fibration* and *orbit spaces* of group actions on manifolds. We give a detailed account of the maps between spaces and divulge exposition outside the realm of the course material where necessary.

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